



Operations & Vector Spaces

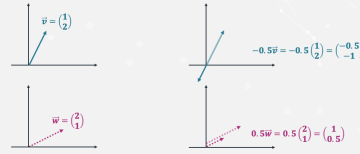
Scalar Multiplication

- When a vector or a matrix is multiplied by a scalar, **each element is multiplied by the scalar**

$$A + B = \begin{bmatrix} 70 & 30 & 15 & 12 & 43 & 0 & 35 & 40 & 32 & 0 & 12 & 46 \\ 45 & 15 & 12 & 43 & 0 & 35 & 40 & 32 & 0 & 12 & 46 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 350 & 150 & 75 & 60 & 225 & 60 & 340 & 175 & 200 & 60 & 230 & 0 \\ 0 & 0 & 215 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

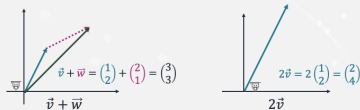
Operations & Vector Spaces

- Multiplying vectors by a scalar is like stretching or **shrinking** the vector



Operations & Vector Spaces

- Adding and scaling vectors create a **vector space**



real numbers, constants or complex numbers used to describe a physical phenomenon with magnitude those representable by a numerical scale, where each specific value indicates a greater or lesser degree on the scale

- Adding and scaling vectors create a vector space

» A **vector space** is defined by:

- Being able to add elements of the space together
- Being able to multiply ("scale") element of the space by numbers called scalars

$$\vec{v} + \vec{w}$$

$$k\vec{v}$$

- » If something is a vector space, you can use linear algebra tools without having to prove anything for that specific vector space

- Three ways to think of **vectors**:

Lists

```
v = [0, 42, 15, 12, 43, 0, 35, 40, 32, 0, 12, 46]
```

Direction & Length



Vector Space

$$\vec{v} + \vec{w}$$

$$k\vec{v}$$

BROADCASTING

Method by which under controlled circumstances you perform operations with tensors of different sizes fills missing positions by repeating the same available data

Although it can be very useful as it condenses and speeds up efficient tensor operation code, it is good practice to track shapes during data manipulation, to help debug problems related to accidental invocation of the Broadcasting mechanism

For example, we can add a column vector with shape (1,2) with a row vector with shape (2,1), by invoking the broadcast mechanism.

```
v = X_train[0].reshape(1, -1)
w = X_train[1].reshape(-1, 1)
print("v = (v)")
print("w = (w)")
print("v + w = (v+w)")
```

```
v = [[16  8]]
w = [[16]
      [8]]
v + w =
[[32  8]
 [16 16]]
```

NORMS

NORM -> THE LENGTH OF A VECTOR

T -> Tail of the vector
H -> Head of the vector

In physics NORM is the vector of length 1, in this case is not the same concept.

VECTOR NORMS IS A WAY TO QUANTIFY VECTOR SIMILARITIES

Measuring Length

- Length is an important feature of vectors

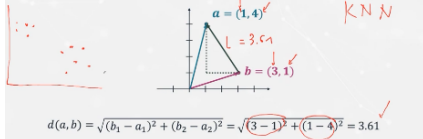
- Length is a measure of distance
- Distance is one way of measuring similarity between things



The first method to calculate similarity between vectors is to compute the length of the resulting vector from their subtraction

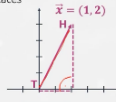
Distance

- Calculating the distance between two points



- What is a formula to calculate the length of a vector?

- The space between places



$$\text{len}(x) = d(T, H) = \sqrt{x_1^2 + x_2^2} = \sqrt{(1)^2 + (2)^2} = \sqrt{5} = 2.27$$

Norms

- Distance formula

$$d(a, b) = \|a - b\|_2 = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + \dots + (b_n - a_n)^2}$$

- Can be used with more than two dimensions

- The **norm** calculates the length of a vector

- The space between places
- Similar to distance calculation

The distance can be expanded to N dimensions, the particular 2 dimensions is called Euclidean distance

Norms

- General **norm** formula:

$$L_p = \|x\|_p = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{\frac{1}{p}}$$

- The L_2 (Euclidean) Norm:

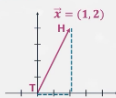
$$L_2 = \|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Distance/Norms

- The most common Norms:

$$L_1 = |x_1| + |x_2| + \dots + |x_n|$$

- Often called the "Manhattan" Norm



The dominant element is L-infinity (can be demonstrated with maths) and is used to find outliers

Distance/Norms

- The most common Norms:

$$L_\infty = \max|x_i|$$

- Used to find extremes or outliers

$$L_1 = (|x_1| + |x_2| + \dots + |x_n|)^{\frac{1}{p}}$$

Distance/Norms

- The most common Norms:

$$L_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

to convert from Euclidean to spherical space

Two vectors or matrices can be multiplied resulting in

- Vectors can be multiplied by each other
 - Rethink $4 \times 5 = 20$ in vector form

$\vec{v} = 4\hat{i}, \vec{w} = 5\hat{i}$

$\Rightarrow 4 \times 5$ is shorthand for adding $\vec{v} = 4$, five times

another vector or matrix of the same size as the origin

Hadamard Product

- Element-wise vector & matrix product
 - Vectors and matrices can be multiplied element-wise when their dimensions are the same

```
arrshape = (3, 5), arrshape = (3, 5)

A =
[[0 1 2 3 4]
 [1 2 3 4 5]
 [2 3 4 5 6]],

B =
[[0 1 0 0 1]
 [0 0 1 0 0]
 [0 0 0 1 1]]

A*B.shape = (3, 5)

A*B =
[[0 1 0 0 4]
 [0 0 3 0 0]
 [0 0 0 6 6]]
```

Other Vector Multiplication Operation

Vector multiplication can also be done with Trigonometry

- Vectors can be multiplied by each other
 - Yes
 - Requires an extra step to get the components of each vector

$\vec{v} \cdot \vec{w} = \vec{v} \cdot \vec{w}_1$

$\vec{v} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \vec{w} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \vec{w}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$

Other Vector Multiplication Operation

- Vectors can be multiplied by each other
 - $\vec{v} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \vec{w} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \vec{w}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$
 - $\vec{v} \cdot \vec{w} = 12$

DOT PRODUCT results in a number, not a structure

Other Vector Multiplication Operation

- Trigonometric properties
 - Calculate length of $\text{len}(\vec{w}_1)$
 - We will use $\cos(\theta) \rightarrow \text{len}(\vec{w}_1) = \text{len}(\vec{w})\cos(\theta)$

$\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$

Other Vector Multiplication Operation

- What is the Dot Product?
 - The Dot Product is the formula created for scalar vector multiplication

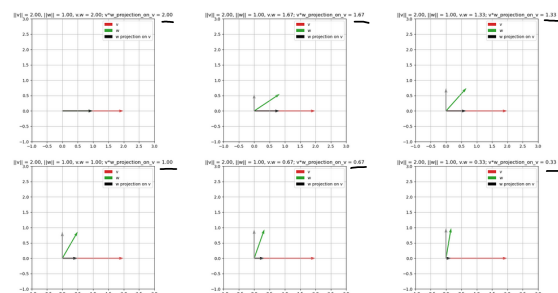
$\vec{v} \cdot \vec{w} = \text{len}(\vec{v}) \cdot \text{len}(\vec{w}_1) = \text{len}(\vec{v}) \cdot \text{len}(\vec{w})\cos(\theta) = \text{len}(\vec{w}) \cdot \text{len}(\vec{v})\cos(\theta)$

$\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos(\theta) = \vec{w} \cdot \vec{v}$

Dot Product

- Cosine Similarity
 - $\cos(\theta) = \frac{\vec{w} \cdot \vec{v}}{\|\vec{w}\| \|\vec{v}\|}$
 - $\cos \theta$ ranges from -1 to 1
 - When $\cos \theta = -1$
 - Vectors are in the exact **opposite direction**
 - When $\cos \theta = 1$
 - Vectors are in the exact **same direction**
 - When $\cos \theta = 0$
 - Vectors are not similar at all (**orthogonal**)

The more the angle grows between vectors, the smaller the projection of one onto the other and therefore the smaller their dot product, making it possible to clearly represent their difference that's why the dot product is used as a tool to calculate similarity between If we keep one vector fixed and increase the angle of the other, the value decreases



Linear Models

- A Linear Model is a dot product between two vectors
- Linear models are additive functions
 - Each term in the function has been "weighted"

$f(x) = x_1 \cdot w_1 + x_2 \cdot w_2 + \dots + x_n \cdot w_n$

weights

Linear Models

- A Linear Model is a dot product between two vectors
- Linear models are additive functions
 - Each term in the function has been "weighted"

$f(x) = x_1 \cdot w_1 + x_2 \cdot w_2 + \dots + x_n \cdot w_n$

$\Rightarrow$ Use the dot product to calculate Linear Models

$f(x) = \vec{x} \cdot \vec{w}$

Estimation Problem

- The Classification problem with Linear Models
 - » Predict whether a visitor to a gardening product page would purchase at least one gardening product during their visit, based on the amount the customer has spent in the product category and their number of months as a Prime member
 - » Change the problem slightly:
 - » Predict how much a person is going to spend on gardening products during their visit

Estimation Problem

- Looking at a new visitor

Visitor tag	Product category	Spend in category	Prime member	Purchased product	Checkout amount
84920394	AB844391	0	0	0	0
84223534	AB844391	70	42	1	156
83202384	AB844391	30	15	0	0
84342229	AB844391	45	12	1	79
8359352	AB844391	0	43	0	0
84332232	AB844391	0	0	0	0
84492902	AB844391	68	95	1	222
84333756	AB844391	0	40	0	0
845452	AB844391	0	32	0	0
8354443	AB844391	0	0	1	44
84302352	AB844391	0	12	0	0
48886763	AB844391	0	46	1	234
83556675	AB844391	57	35	1	255
84200532	AB844391	0	24	0	0
84228325	AB844391	73	41	1	204
8341421	AB844391	0	23	0	0
84567549	AB844391	0	9	1	15

$$\vec{x} = \begin{pmatrix} 63 \\ 33 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} 2.5 \\ 0.8 \end{pmatrix}$$

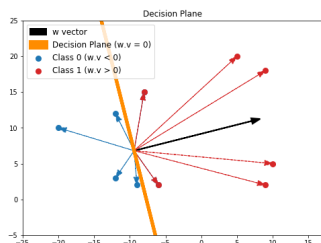
$$63 \cdot 2.5 + 33 \cdot 0.8 \approx \$186$$

The simplest Linear Model using the dot product between vectors is the Decision Plane
 forma más simple (plano ortogonal) permite decidir entre dos opciones creando un Plano vectorial que separa ambas
 It's about finding the plane line where multiplying X by weight W if > 0 will be
 in this case it is BUY or NO_BUY

Decision Plane

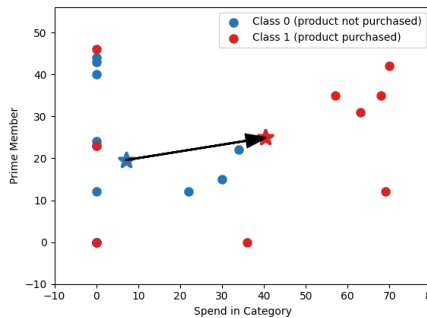
This storyline gives us an even easier algorithm to make class predictions from just the sign of dot products. For example, to solve a binary classification task, we would pick a vector w pointing towards say, class 1; all the (training data points) vectors v such that $w \cdot v > 0$ would be pointing in roughly the same direction as w , and thus be assigned to class 1; whereas all the vectors v such that $w \cdot v < 0$ would be pointing in the opposite direction, and thus be assigned to the other class, class 0.

Most importantly, from a geometric point of view, this is also to say that, given a vector w - the black vector in the figure below, all vectors v such that the dot product $v \cdot w = 0$, are orthogonal on the vector w . The heads of all these vectors v determine a so-called hyperplane orthogonal on the vector w - here a line, the orange line depicted below. Furthermore, all vectors v such that $v \cdot w > 0$ are pretty much pointing in the same general direction with the vector w , on one side of the hyperplane (the red vectors depicted below), whereas all vectors v such that $v \cdot w < 0$ are pretty much pointing in the opposite direction of the vector w , on the other side of the hyperplane (the blue vectors depicted below). And so, this hyperplane described by a dot product, splits the data space into two subspaces, each subspace corresponding to distinct classes associated to some particular values of the dot product with respect to a threshold. We can think of this hyperplane as a binary classifier that separates the two classes. In this context, such hyperplane is often referred to as a *decision plane*.



In summary, the steps to generate a binary decision plane are

* Computar la media de los vectores `ave_0` y `ave_1` asociados a las clases 0 y 1 respectivamente con el training dataset X_{train}
 Usando the resultant vector W connecting both class means, compute the dot product with the entire X_{test} dataset
 Make predictions on X_{test} , trying different θ , nacerito sobre $1, 2, 3, -1, -2, -3, \dots$ sobre el producto vectorial recogiendo cual de ellos da mejor rendimiento



Se puede observar que prácticamente todos los puntos rojos están en la misma dirección que el vector W (producto escalar positivo) mientras que la mayoría de los azules tienen opposite direction

It is possible that threshold 0 does not give us the best decision plane orthogonal to the weight vector but different threshold choices which correspond visually to moving the line orthogonal to the left or to the right improve the classifier
 We would need to iterate to find the threshold value so instead of greater/less than 0, do it with etc,

KNN vs Decision Plane for binary clasification

When evaluating models like KNN (non-parametric, learns nothing, calculates all distances) vs decision plane (where we learn because we take the mean of all points + threshold, not needing to pass through all points again), there are typically 3 evaluation methods:

Mean of absolute distances between value and prediction
 don't appear, it must be squared

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

Square root so the distance is understandable, since the previous must be squared to avoid negative distances

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

Does not give absolute distance but how much variation is explained by the model

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$